

Class 6: Files, pandas, and Plotting

Data-driven Systems Engineering

Agenda

- File handling in Python: opening, reading, writing.
- Why pandas exists: the DataFrame and Series structures.
- Reading tabular data into a DataFrame.
- Selecting data with `loc`, `iloc`, and boolean masks.
- A first look at plotting with pandas and Matplotlib.

Learning goals

- Open a file in the correct mode for the task at hand.
- Explain the difference between a pandas Series and a DataFrame.
- Load a real dataset into a DataFrame with `pd.read_csv`.
- Select rows and columns with `loc`, `iloc`, and a boolean condition.
- Produce a first plot directly from a DataFrame.

Course Map and Running Cases



Today in the course

Focus: Getting data in and out of Python, and a first look at it visually.

Bridge: Variables and loops manipulate values already in memory; this class is about where those values come from and how we glance at them.

Fraud case

The fraud model is trained from a CSV pulled straight into a DataFrame, and its artifacts are written and read back as pickle files on disk.

Course thread: Nothing downstream — EDA, modeling, serving — can happen until data reliably gets in and out of the program.

Pasteurization case

The pasteurization case writes and rereads sensor batches the same way, before any streaming or dashboard logic gets involved.

File handling in Python

- `open(filename, mode)` is the core function for reading, writing, and appending files.
- Always close what you open, or use `with open(...) as f:` so Python closes it automatically.

Mode	Meaning
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<code>r</code>	Read (default); errors if the file does not exist.
<code>w</code>	Write; creates the file, overwrites if it exists.
<code>a</code>	Append; creates the file if needed, adds to the end otherwise.
<code>x</code>	Create; errors if the file already exists.
<code>t / b</code>	Text mode (default) or binary mode, e.g. for pickled objects.

Writing a file, for real

```
with open('model.pkl', 'wb') as f:  
    pickle.dump(cls, f)  
  
with open('scaler.pkl', 'wb') as f:  
    pickle.dump(scaler, f)
```

- From 'streamlit/modeling.py': mode 'wb' combines write and binary, because a pickled Python object is not text.
- The trained model and scaler exist on disk only because of these two `open()` calls.

Reading it back

```
model = pickle.load(open('model.pkl', 'rb'))  
scaler = pickle.load(open('scaler.pkl', 'rb'))
```

- 'App_Credit_Card.py' reopens the same two files in 'rb' mode — read, binary.
- Whoever writes a file and whoever reads it must agree on the mode; mixing text and binary modes is a common source of errors.

Why pandas

- pandas is a fast, flexible library for tabular data analysis; many ML libraries accept its structures directly as input.
- **Series**: a single labeled column of data.
- **DataFrame**: a table made of one or more Series sharing a row index, similar to a spreadsheet or a relational table.
- pandas is column-oriented: operations on a whole column are typically fast and expressive.

Reading tabular data into a DataFrame

```
import pandas as pd

url = 'https://storage.googleapis.com/download.tensorflow.org/data/creditcard.csv'
df = pd.read_csv(url)
```

- This is the exact loading line from 'streamlit/modeling.py': `read_csv` works the same whether the source is a local path or a URL.
- One function call replaces the manual file-opening and parsing you would need without pandas.

A pandas cheat sheet

Call	What it shows
<code>df.head()</code>	The first few rows, for a quick sanity check.
<code>df.info()</code>	Column names, types, and non-null counts.
<code>df.describe()</code>	Summary statistics for numeric columns.
<code>df.shape</code>	A (rows, columns) tuple.
<code>df.columns</code>	The list of column names.

Selecting rows and columns

```
X = df.drop('is_fraud', axis=1)    # every column except the target  
y = df['is_fraud']                # a single column, as a Series  
  
fraud_only = df[df['is_fraud'] == 1] # boolean mask on rows
```

- `df['col']` selects a column by label; `df.drop(col, axis=1)` selects everything else.
- `df.loc[...]` selects by label, `df.iloc[...]` selects by integer position, and a boolean expression inside `df[...]` filters rows.
- The column name `is_fraud` is the renamed target from 'modeling.py'; the boolean mask above is the standard pandas way to isolate the fraud cases.

Plotting straight from pandas

<code>kind=</code>	Produces
<code>'line'</code>	Line plot (the default).
<code>'bar' / 'barh'</code>	Vertical or horizontal bar plot.
<code>'hist'</code>	Histogram.
<code>'box'</code>	Boxplot.
<code>'scatter'</code>	Scatter plot.

- A DataFrame's `.plot()` method calls Matplotlib under the hood — one line, no separate import needed for the simplest cases.

Plotting example

```
iris.groupby("species").mean().plot(kind="bar", figsize=(10, 6))  
plt.title("Mean Feature Values by Species")  
plt.ylabel("Mean Value")  
plt.show()
```

- From 'notebooks/Class2_EDA.ipynb': group first, then plot the grouped result directly.
- `plt.title` and `plt.ylabel` come from Matplotlib and layer on top of the pandas-generated plot.
- Class 10 returns to this notebook for a full exploratory analysis; today is only the mechanics of getting a plot on screen.

How the repo already uses today's building blocks

Repo activity

- 'streamlit/modeling.py' reads a CSV straight from a URL into a DataFrame with one call.
- 'streamlit/modeling.py' and 'App_Credit_Card.py' write and read the same two pickle files in matching binary modes.
- 'notebooks/Class2_EDA.ipynb' shows the shortest path from a DataFrame to a plot.

Question

'modeling.py' loads its dataset from a URL every time it runs, with no local caching. What could go wrong with that in a course lab, or later in a production pipeline?

- Surfaces reproducibility and availability concerns without yet naming them as “data engineering.”

Papers and materials

[pandas documentation](#); [pandas overview](#); [Matplotlib documentation](#).

From This Class to the Next

What stays with us

- File modes decide whether data is read, written, appended, and as text or binary.
- DataFrames and Series give tabular data a shape Python can manipulate efficiently.
- Selection with `loc`, `iloc`, and boolean masks is the everyday way to carve out the rows and columns you need.

Project use

Any project that reads a dataset or saves a model artifact is already using today's material, whether or not it is named explicitly.

Next connection

Class 7 moves from data mechanics to code quality: functions, modules, and the good practices that keep a growing script readable.

Course thread: Getting data in and out cleanly is only useful if the code around it stays maintainable — that is next.